

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/258835904>

Review of echolocation in insectivores and rodents

Chapter · January 2004

CITATIONS

14

READS

2,506

2 authors, including:



[Jeanette Anne Thomas](#)

Western Illinois University

130 PUBLICATIONS 3,243 CITATIONS

SEE PROFILE

calized during an experiment on "hole finding." This contradicts our 1997 and 1998 observations of Weddell seals, in which all the seals observed to vocalize did so during ascent to their breathing hole and increased their repetition rate as distance to the hole decreased. Of the 34 types of Weddell seal sounds from McMurdo, all sounds produced by instrumented, diving seals while approaching the breathing hole were low in frequency, repeated in a series, and had an accelerating repetition rate as the seal ascended to a hole. The seals in the studies by Wartzok et al. had acoustic transmitters attached to the their backs. Although ultrasonic (60–70 kHz, source level of 159 dB re 1 μ Pa), the transmitters were probably audible to the seals wearing them. At 65 kHz, this source level would be 45 dB down from the ringed seal's maximum sensitivity at 32 kHz (Wartzok et al. 1992b). A signal at effectively 114 dB re 1 μ Pa should be audible to the seal. Furthermore, the rate of pulsing (every 1.5 s) was modulated by depth, so it changed as the seal dove. We wonder whether the changing pulsing sound could have inhibited the seal's vocal behavior, thus obscuring the researcher's ability to examine the possible use of sounds for orientation. The video-data logger system used in our study did not use a pinger and so was relatively quiet (except for low level motor noise from the camera).

Why was prey capture by the Weddell seals not always associated with sounds? Why were only two instrumented seals quite vocal? These are good questions that should be addressed. Perhaps there is a "social etiquette" dictating when echolocation-like signals should be used, such that not all seals vocalize; yet all seals hear and use the information. It makes sense that if a colony of 50–100 seals all produced sound signals at the same time, it might be difficult to evaluate the reflections. The

idea of passive listening to echolocation sounds from other animals has a foundation in Xitco and Roitblat's (1996) bottlenose dolphin studies. They demonstrated that a neighbor dolphin could perform an echolocation detection task simply by listening to the test animal's signal.

Lastly, there is the notion that echolocation signals are not restricted to only that purpose. There could be a gradient from active use of sounds for only navigation, to dual use of some sounds for navigation and communication, to the passive use of typical communication sounds for orientation cues as a supplement to vision. It is not impossible that seals vocalizing while ascending to their breathing holes are warning others ("I am coming, get out of my hole") with additional information as to how far away they are. The Weddell seal, and other polar pinnipeds, pose many unanswered questions related to the use of sound for navigation or foraging.

There is no positive evidence that seals, including Weddell seals, use echolocation. It is interesting to note, however, that some seals do have all the acoustic tools necessary for such behavior. This preliminary study demonstrates the value of studying these behaviors from instrumentation carried by the seal.

Acknowledgments

We acknowledge the support of the National Science Foundations, Polar Program, grant OPP9614857, and the National Undersea Program, grant UAF 98-0040. We also acknowledge the individuals who participated in obtaining data in the field: T. Williams, L. Fuiman, M. Horning, W. Hagey, S. Kanatous, S. Cohin, D. Calkins, S. Collier, and R. Skrovan.

{ 72 }

Echolocation in Insectivores and Rodents

Jeanette A. Thomas and Mersedeh S. Jalili

Introduction

In the 1950s and 1960s, development of equipment capable of sensing, recording, and analyzing ultrasonic sounds facilitated the discovery of echolocation abilities in chiropterans (bats) and odontocetes (toothed whales and dolphins). Echolocation aids navigation in visually limited environments or in capture of fast-moving prey.

For example, echolocation is particularly useful for nocturnal, insectivorous bats that need to coordinate their flight pattern relative to moving insect prey. Echo ranging is adaptive for dolphins that need to swim quickly and accurately to capture fast-moving fish at dark oceanic depths. The emphasis of echolocation studies on these two groups of mammals perhaps was driven by a special fascination for them or availability of funds for

modeling animal echolocation for the improvement of humanmade sonars. Only a few investigators examined the potential of echolocation in other mammals (Airapet'yants and Konstantinov 1970; Airapet'yants 1974; Sales and Pye 1974). Clark and Ellison (chapter 73, this volume) discuss the use of acoustic cues for orientation by baleen whales. Schusterman et al. (chapter 69, this volume) summarize studies on possible echolocation in seals and sea lions. Herein, we summarize the few studies on echolocation abilities in insectivores and rodents and provide some new data for each group.

According to Wilson and Reeder (1993), the six extant families of Insectivora include Soricidae (shrews), Erinaceidae (hedgehogs), Chrysochloridae (golden moles), Solenodontidae (solenodonts), Tenrecidae (tenrecs), and Talpidae (moles and desmans). Sometimes echolocation aids in pursuit of fast-moving prey, but prey

sought by insectivores are not particularly quick moving (e.g., ground insects, mollusks, worms, grubs, insect larvae) and insectivores do not stalk them over great distances. However, echolocation also is useful in environments with limited light, and many insectivores live in dark environments (table 72.1). For example, moles and golden moles are totally fossorial and feed, move, and raise young exclusively under ground. Shrews are nocturnal, nest in underground burrows, and move within dense herbaceous ground cover, runways under humus, or runways made by small rodents (George, Choat, and Genoways 1986; Whitaker 1974).

Wilson and Reeder (1993) recognized 29 families of rodents, constituting 43% of all mammalian species. Like insectivores, rodents generally do not hunt fast-moving prey; many are, in fact, herbivorous or granivorous. Nevertheless, nocturnal, semifossorial, or fossorial

TABLE 72.1. Summary of diet and geographic distribution in insectivores. (Data taken from Nowak 1991.)

Family	Subfamily/Genera	Habit	Diet	Distribution
Talpidae	15 genera, 32 species			
	Subfamily: Talpinae (Old World moles, <i>Talpa</i>)	fossorial	worms, insect larvae	Eurasia
	Subfamily: Scalopinae (New World moles, <i>Scalopus</i> , <i>Neurotrichus</i> , <i>Parascalopus</i> , <i>Scapanus</i>)	fossorial	worms, insect larvae	North America
	Subfamily: Condylurinae (star nose moles, <i>Condylura</i>)	fossorial	aquatic insects, small fish, crustaceans	North America
	Subfamily: Desmaninae (<i>Desmana</i> and <i>Galemys</i>)	semiaquatic	small fish, crustaceans, insects, worms	Russia; Spain and Portugal
Tenrecidae	10 genera, 23 species			
	<i>Potamogale</i> 1 species	semiaquatic	crabs, fish, amphibians	Central Africa
	<i>Micropotamogale</i> 2 species	semiaquatic	worms, insects, crabs, fish, frogs	Guinea, Liberia, Zaire
	<i>Microgale</i> 11 species	terrestrial and semiaquatic	insects	Madagascar
	<i>Hemicentetes</i> 1 species	terrestrial	earthworms	Madagascar
Chrysochloridae	7 genera, 18 species	fossorial	worms, insects, invertebrates	Africa, Cameroon/ Uganda south to Cape Good Hope
Solenodontidae	1 genus, 2 species <i>Solenodon</i>	semifossorial	invertebrates, reptiles, vegetation	Cuba and Hispaniola
Soricidae	22 genera, 289 species			
	Subfamily: Soricinae (<i>Sorex</i> , <i>Cryptotis</i> , <i>Blarina</i> , <i>Notiosorex</i> , <i>Megasorex</i> , <i>Neomys</i> , <i>Nesiotites</i> , <i>Soriculus</i> , <i>Nectogale</i> , <i>Chimarrogale</i> , <i>Anourosorex</i>)	terrestrial	invertebrates, insects	North America, South America, Europe, Asia, Africa
	Subfamily: Crocidurinae (<i>Crociodura</i> , <i>Paracrocidura</i> , <i>Suncus</i> , <i>Sulvisorex</i> , <i>Ruwenzorisorex</i> , <i>Feroculus</i> , <i>Solisorex</i> , <i>Myosorex</i> , <i>Diplomesodon</i> , <i>Scutisorex</i>)	terrestrial	invertebrates, insects	North America, South America, Europe, Asia, Africa
Erinaceidae	9 genera, 20 species			
	Subfamily: Galericinae (gymnures, moonrats)	terrestrial	invertebrates, carrion	Africa, Europe, Asia
	Subfamily: Erinaceinae (hedgehogs)	terrestrial	invertebrates, insects	Africa, Europe, Asia

TABLE 72.2. Summary of diet and geographic distribution in fossorial rodents. (Data taken from Nowak 1991.)

Family	Genera	Habit	Diet	Distribution
Muridae	<i>Nannospalax</i> 3 species	fossorial	tubers, roots, underground vegetation	Balkan Peninsula, Asia Minor, Syria, Palestine, Egypt, Libya
Muridae	<i>Spalax</i> 5 species	fossorial	tubers, roots, bulbs	Romania, Ukraine
Muridae	<i>Myospalax</i> 6 species	fossorial	roots, grains	China
Muridae	<i>Tachyrorvctes</i> 2 species	fossorial	roots, rhizomes, tubers, bulbs	Ethiopia, Somalia, Uganda, Rwanda, Kenya, Tanzania
Bathyergidae	<i>Georchys</i> 1 species	fossorial	tubers, roots, bulbs	Cape of Good Hope, South Africa
Bathyergidae	<i>Cryptomys</i> 3 species	fossorial	roots, tubers, invertebrates	western Zambia to northwestern South Africa
Bathyergidae	<i>Heliophobius</i> 1 species	fossorial	tubers, bulbs	southern Kenya, Tanzania, southeast Zaire, eastern Zambia, and Mozambique
Bathyergidae	<i>Bathyergus</i> 2 species	fossorial	bulbs, roots	Namibia and southern South Africa
Bathyergidae	<i>Heterocephalus</i> 1 species	fossorial	tubers, roots	eastern Ethiopia, central Somalia, and Kenya
Ctenomyidae	<i>Ctenomys</i> 44 species	fossorial	roots, stems, grass	South America
Geomyidae	6 genera, 37 species	fossorial	roots, tubers, stems	North America, Canada south to Colombia

rodents could benefit from extrasensory inputs, such as echolocation (table 72.2).

Bats and rodents evolved directly from the most primitive placental mammal, the insectivores (Alderton 1996). We suggest that echolocation is a primitive characteristic in mammals, perhaps evolving as early as in insectivores, then further refined by bats and perhaps some rodents (see discussion in Denzinger, Kalko, and Jones, chapter 42, this volume). Clearly, more research is needed on the possible echolocation abilities in insectivores and rodents.

From a search of the literature, it appears that certain adaptations or environmental conditions are related to echolocation abilities. Below, we describe these factors for the known echolocators, such as bats and dolphins, and discuss these characteristics in some insectivores and rodents.

LIMITED AMBIENT LIGHT

Most echolocating mammals live at least part of their life in darkness or low light environments. For example, most bats are nocturnal and maneuver easily in dark caves or crevices. Ambient light for toothed whales and dolphins is limited during nighttime foraging, in deep water, turbid water, ice-covered areas, or amid a plankton bloom. Some insectivores and rodents are active mainly at night (Whitaker 1974), creating the need to orient and forage in limited environmental light. For semifossorial or fossorial mammals that nest in underground burrows, forage in underground tunnels, or move in dense herbaceous covered runways, ambient light is limited at all times.

Feldhamer et al. (1999) identified groups of mammals that are fossorial or semifossorial: insectivore families Soricidae, Talpidae, Solenodontidae, Chrysochloridae; rodent families Geomyidae (pocket gophers), Ctenomyidae (tuco tucos), Bathyergidae (blesmois), and Mu-

ridae (subfamilies Myosplacinae, East Asian mole-rats; Spalacinae, blind mole-rats; and Rhizomyinae, African mole-rats); a xenarthra family, Dasyuridae (armadillos); and a marsupial family, Notoryctidae (marsupial moles). Herein, we only discuss possible echolocation in fossorial insectivores and rodents. (Marsupial moles are not discussed; however, they also have characteristics reminiscent of echolocation—e.g., fossorial, small ear opening, vestigial eyes, no lens or pupil, reduced optic nerve, diet of earthworms and insect larvae, production of low-intensity sharp squeaks [Nowak 1991]. If echolocation occurred in this family, it would represent a separate evolution from placental mammals.)

LIMITED VISION

Bats

Mammals with poorly developed vision must rely on other senses, like olfaction, tactition, passive acoustics, or echolocation for orientation or foraging. In bats, echolocation is the predominant sensory mode. Anatomically, bat vision is not limited. In fact, "blind as a bat" is only an anecdotal description (Fenton 1983).

Odontocetes

In odontocetes, acoustics (passive listening or active echolocation) is the predominant sensory channel. Eyes of toothed whales are compromised anatomically for vision both under water and in air, but they see well over short distances in either environment (1999). River dolphins are the exception, being essentially blind from a reduced visual anatomy.

Insectivores

All insectivores have small eyes and probably poor vision (table 72.3). The optic region in the shrew brain is small and poorly developed. Churchfield (1990) described shrews as having underdeveloped eyes, only 0.7–

TABLE 72.3. Summary of sensory anatomy in insectivores. (Data taken from Nowak 1991.)

Family	Subfamily/ Genera	Eyes	Ears	Vibrissae	Sounds
Talpidae	Subfamily: Talpinae	minute, hidden in fur; only detect light	no external ear	keen sense of vibration; odiferous, long snout	no information given
	Subfamily: Scalopinae	minute, hidden in fur	no external ear	keen sense of vibration	<i>Neuortrichus</i> taps nose to hunt; faint audible twitter
	Subfamily: Condylurinae	minute, visible	no external ear	nose has bare tentacles that move when feeding; odiferous	no information given
	Subfamily: Desmaninae	small, poor vision	small external ear	long snout, scent glands	<i>Galemys</i> echolocates (Richard 1973)
Tenrecidae	<i>Potamogale</i>	minute	external ear present	present	no information given
	<i>Micropotamogale</i>	minute	external ear present	present	no information given
	<i>Microgale</i>	minute	prominent external ear	present	<i>M. dobsoni</i> echolocates (Gould 1965)
	<i>Hemicentetes</i>	minute	external ear present	present	audible "putt-putt" and "crunch"; echolocates; Gould 1965
Chrysochloridae	<i>Fremitalpa</i>	eyes fused at early age; skin thickens over eye with age; lacks temporal bullae	small, hidden in fur	long snout, leathery nose	no information given
	<i>Cryptochloris</i> and <i>Chrysochloris</i>	eyes vestigial, skin-covered; temporal bullae present	small, hidden in fur	long snout, leathery nose	no information given
	Other species	eyes vestigial and covered	small, hidden in fur	long snout, leathery nose, with hairy skin	no information given
Solenodontidae	<i>Solenodon</i>	small	external ear present	present	high-frequency clicks, ~31 kHz, 0.1–3.6ms; perhaps echolocation
Soricidae	Subfamily: Soricinae	small, poor vision	small ears hidden in fur, keen hearing	many present, keen smell, odiferous	suspected echolocation
	Subfamily: Crocidurinae	small, poor vision	small ears hidden in fur	many present	no evidence of echolocation
Erinaceidae	Subfamily: Galericinae	small, poor vision	no information	many present	no information
	Subfamily: Eriaceinae	small, poor vision	keen hearing	many present	hiss, scream

1.5 mm in diameter (figure 72.1), and recognized three categories of eyes in insectivores: small, but visible among the fur (in terrestrial and scansorial species); extremely small and obscured by the fur (in psammophilic and semifossorial species); and totally covered by skin (in semiaquatic species). The primary function of insectivore eyes could be in sensing shades of light and dark. George, Choat, and Genoways (1986) described ocular

structure and function in the short-tailed shrew (*Blarina brevicauda*) as "degenerate": "the optic nerve is diminutive and only slight motion is possible, ocular muscles do not rise directly from the skull, vision probably is limited to perception of light, the lachrymal gland is larger than the eyeball and secretes tears to protect and wash away dirt from the eye" (p. 4). The fossorial coast mole, *Scapanus orarius*, has small eyes, the auditory meatus is



Fig. 72.1. Representatives of the two genera of Soricidae. *Top:* *Blarina brevicauda* (photograph by R. Altig). *Left:* *Cryptotis parva* (photograph by F. A. Cervantes Reza).

concealed by fur, and the animal is primarily nocturnal (Hartman and Yates 1985). The talpid, shrew-mole (*Neurotrichus gibbsii*) is the only vertebrate to have a pigmentation layer covering the surface of the lens of the eye.

Churchfield (1990) held that the sense of touch plays an important role in shrew orientation—"judging the size of holes and crevices through which it squeezes or the nearness of objects in its path." She described shrew snouts as "long with many vibrissae" and noted that they "constantly wiggle to and fro, sniffing the air and probing the substratum" (p. 18). Moles have specialized touch-sensitive, Eimer's organs on their snouts (Church-

field 1990). The talpid, shrew-mole has eight pairs of vibrissae on the snout and locates prey by tapping its nose on the ground directly in line with the prey, then 40 degrees to one side, then the other. Finally, the animal moves one step forward and continues this process until its nose touches the prey. The semiaquatic, nocturnal Iberian desman (*Galemys pyrenaicus*) has Eimer's organs dispersed among the vibrissae that are believed to play a role in orientation and detection of prey (Palmeirim and Hoffmann 1983). It actively explores the bottom of river streams, moving sediment with the snout, head, and forefeet. The Iberian desman essentially is

TABLE 72.4. Summary of sensory anatomy in fossorial rodents. (Data taken from Nowak 1991 and Rado et al. 1987.)

Family	Genera	Eyes	Ears	Vibrissae	Sounds
Muridae	<i>Nannospalax</i>	no eye opening	reduced to low ridge	facial bristles	during mating tap head against burrow
Muridae	<i>Spalax</i>	no eye opening	reduced to low ridge; keen hearing	line bristles aid navigation; poor olfaction	recognize objects by touch
Muridae	<i>Myospalax</i>	small, hidden in fur; sensitive to light	no external ear	few short whiskers	squeal when frightened
Muridae	<i>Tachyoryctes</i>	small, visible, functional	small ears	stiff hairs on face tactile role	no information given
Bathyergidae	<i>Georychus</i>	minute	small opening surrounded by skin	present	males drum hind feet during breeding
Bathyergidae	<i>Cryptomys</i>	very small cornea; sensitive to air current	very small; sensitive to sound or vibration	present	squeals, grunts, growls
Bathyergidae	<i>Heliophobius</i>	extremely reduced eyes, unreceptive to light	reduced external ears; keen hearing	present	no information given
Bathyergidae	<i>Bathyergus</i>	small	no external ear	present	chattering noise while burrowing
Bathyergidae	<i>Heterocephalus</i>	minute	small external ear	nearly naked, lip vibrissae	no information; eusocial
Geomyidae	6 genera, 37 species	small	small external ear	present	squeals and hisses
Ctenomyidae	<i>Ctenomys</i>	small	external ear reduced, auditory bullae enlarged	present; tail sensory hairs	"tloc-tloc-tloc" by males during breeding, 10–20 s, accelerating rate

silent, but Richard (1973) reported aggressive sounds in captive Iberian desmans. Richard (1973) speculated desmans use sounds from the slapping of the forefeet against the water for echolocation.

The sense of smell in long-snouted insectivores also plays an important role in orientation and foraging. Many insectivores are notably odiferous. Some species have scent glands and use them in interspecific contexts (Churchfield 1990).

Rodents

In rodents, both hearing and smell are vitally important senses. Rodents active during daylight are more reliant on vision, whereas rodents active at night or living under ground depend less on sight (Alderton 1996). Nocturnal rodents, like rats and flying squirrels, generally have relatively large eyes to compensate for low light levels. Fossorial rodents have small eyes, indicating their uselessness under ground (table 72.4). In the family Bathyergidae, two species of African mole rats, *Bathyergus* spp., occur in sandy soils in hot dry regions of Africa, burrow extensively with complex tunnels nearly 300 m long, and feed on underground bulbs and tubers (1999). Several mole-rat species (eight in *Cryptomys*, two in *Georychus*, one in *Heliophobius*, and one in *Heterocephalus*) have minute eyes concealed beneath the skin and are entirely blind; they have small pinnae and their auditory acuity is reduced in favor of enhanced tactile and olfactory sensitivity. Similarly reduced or ob-

scured eyes, long vibrissae, and small ears characterize some muridae (e.g., *Spalax*, *Nannospalax*, *Myospalax*, and *Tachyoryctes*; Feldhamer et al. 1999). In these fossorial murids, animals recognize objects by touching with the nose. Tactile and auditory senses are acute, but olfaction is considered weak (Feldhamer et al. 1999).

BROADBAND HEARING

Broadband, ultrasonic hearing often (but not always) is associated with echolocation (Airapet'yants and Konstantinov 1970). Both bats and dolphins typically have ultrasonic hearing (see Vater and Koessl, chapter 12, this volume). Using a short, broadband echolocation signal (as in both bats and dolphins) allows rapid sampling of target details across a range of frequencies.

Hearing in insectivores has not been tested behaviorally; however, anatomical data by Henson (1961) suggests that shrews have ultrasonic hearing up to 50 kHz. However, he believed that moles probably are unable to detect ultrasounds. Broadband or ultrasonic hearing is expected in insectivorous mammals with small head sizes. Airapet'yants (1974) used conditioned reflex techniques in hedgehogs (family Erinaceidae) to show they hear up to 45 kHz. Shrews have a unique hearing apparatus in that the auditory bullae are completely unossified and appear as an open hole on the underside of the skull. The significance of these open bullae is unknown. The temporal bullae are present in some moles and golden moles, but not all (table 72.4).

The prominence of an external ear in shrews varies with habit and habitat. Scansorial, or ground-dwelling, species have the largest pinnae. Temperate species have quite large hairy ears, while Old World tropical species have large, nearly naked ears. Semifossorial shrews have small pinnae, often obscured by fur. Some aquatic shrews lack pinnae entirely (Churchfield 1990).

Hearing has been studied extensively in rodents (Airapet'yants and Konstantinov 1970; Fay and Popper 1994), especially in chinchillas (*Chinchilla laniger*), laboratory rats (*Rattus norvegicus*), gerbils (*Meriones unguiculatus*), house mice (*Mus musculus*), and guinea pigs (*Cavia porcellus*). Of these species, all showed ultrasonic hearing. The golden hamster (*Mesocricetus* spp.) hears up to 23 kHz. Guinea pigs and redbacked voles (*Clethrionomys* spp.) hear up to 50 kHz. Laboratory rats, garden dormice (*Eliomys* spp.), and harvest mice (*Reithrodontomys* spp.) hear up to 60 kHz. The house mouse hears up to 100 kHz. In many species there are two regions of sensitivity, one sonic and one ultrasonic. For example, laboratory mice were sensitive from 4 to 10 kHz and again at 40 to 45 kHz, and redbacked voles were sensitive at 13 kHz and 50 kHz (Airapet'yants 1974). Kahmann and Ostermann (1951) acknowledged the possibility that besides functioning in social contexts, ultrasonics in rodents could serve in echolocation.

ULTRASONIC SOUND PRODUCTION

Echolocating mammals often produce ultrasonic sounds, either in a series of frequency-modulated (FM) sweeps or in a train of pulses. Repetition rate of sounds increases as the sender approaches a target (i.e., prey or object in the environment). The short wavelength of high-frequency signals provides good details of targets. However, mammalian production of ultrasonic vocalizations alone is not evidence for echolocation, because some ultrasonics are used for communication.

Hamilton (1940) observed that smokey shrews "when foraging for food . . . utter an almost indiscernible twitter . . . while rooting through the leaf mold or appearing on the forest litter about rotted logs, the twitching nose and vibrissae held aloft and this faint, almost inaudible twittering was kept up continually." Churchfield (1990) reported that shrews are very vocal and emit a wide range of sounds, including ultrasonics. She categorized shrew sounds into calls of alarm, defense, and aggression; calls during courtship; sounds associated with mother-young interactions; and sounds produced during exploration and foraging. High-frequency "twitters" and other sounds are produced by shrews as they explore novel situations. They emit these sounds less often as they become familiar with their environment (Churchfield 1990). Unlike rodents, shrews do not use ultrasound in the nest for mother-infant communication (Churchfield 1990). Shrews produce sounds in three ways: "puts" and "hisses" coupled with exhalation and

inhalation through the nose; "clicks" produced by the tongue; and "chirps," "twitters," and "shrieks" produced by the larynx (Churchfield 1990).

Kahmann and Ostermann (1951) noted that high-frequency sounds in small mammals could be used for acoustic location. Several rodents in the family Muridae produce a variety of ultrasonic sounds for mother-young communication or as distress calls (Airapet'yants 1974; Bell 1979; Elwood and Keeling 1981; Newman 1988; Blake 1992). In these contexts, the need to vocalize for interspecific communication must be balanced against the costs of increased detection by a predator (Sales and Pye 1974; Alderton 1996). Blumberg and Alberts (1990) suggested that the ultrasonic components of infant rodents are only a by-product of respiration; however, Blake (1992) disputed this.

Some insectivores and rodent have similar characteristics correlated with echolocation in bats and dolphins. Species that fit this profile need to be identified and studied for possible echolocation abilities.

Previous Studies on Echolocation in Insectivores

Sales and Pye (1974), Buchler (1976), Buchler and Mitz (1980), and Gould (1980) proposed a variety of echolocation abilities among insectivore groups. Most information on insectivore acoustic abilities is reported in the family Soricidae (22 genera, approximately 322 species). Species in this family are thick-bodied, robust animals able to push their way through soil and leaf litter using the long snout, teeth, and forefeet to excavate a passage. Shrews have small eyes and pinnae obscured by the fur (fig. 72.1, table 72.3). They use high-frequency sounds for intraspecific communication, orientation, and prey detection (Feldhamer et al. 1999). Shrews eat mainly earthworms and soil-dwelling insect larvae.

Studies by Gould, Negus, and Novick (1964), Gould (1969), Buchler (1976), Tomasi (1979), and Forsman and Malmquist (1988) suggested that only shrews of the subfamily Soricinae echolocate. Gould (Airapet'yants 1974, reference 523) first studied possible echolocation in shrews by having them run an obstacle course of glass bubbles. The lightest touch of a bubble would cause it to drop, signifying the shrew ran into it. Intact shrews and shrews with no vibrissae ran the course with few collisions; however, shrews with plugged auditory canals ran the course more slowly and made more errors. Because this technique was imprecise, Gould, Negus, and Novick (1964) tested possible echolocation in *Sorex palustris*, *S. vagrans*, and *S. cinereus* using a disk-platform apparatus. To receive a reward, the shrew entered a vinyl tube, exited onto a disk platform, and jumped down to a runway containing a mealworm and water. Shrews were observed in the dark with an infrared "snoop scope." Non-auditory cues, such as smell or touch, were eliminated. After a learning period, performance of all shrews was high. However, when the ears of shrews were plugged,

TABLE 72.5. Pulse characteristics of several shrew species.

Species	Pulse mean frequency	Short pulse mean duration	Long pulse mean duration
<i>Sorex vagrans</i>	37 kHz	5.3 ms	18.0 ms
<i>Sorex palustris</i>	36 kHz	5.1 ms	18.7 ms
<i>Blarina brevicauda</i>	42 kHz	2.3–7.0 ms	20.0 ms

performance dropped to chance. Initially, shrews produced low-amplitude, short pulses, but once the disk platform was recognized they produced higher amplitude, longer duration pulses. Typical pulses were 0.02–0.14 μ bar, 4.9–33.0 ms, and 30–60 kHz (Sales and Pye 1974). Pulses were not repeated often, only up to 13–16 pulses in an unfamiliar situation. Table 72.5 lists the characteristics of pulses by species.

Buchler (1976) conducted studies on the performance of wandering shrews (*Sorex vagrans*) using a similar disk platform with microphones surrounding the apparatus. He noted the same high performance for normal shrews, but decreased performance in hearing-impaired shrews.

Using a Y-maze, Buchler tested the wandering shrew's ability to detect a plate blocking one arm of the maze and then choose the other, open-arm of the maze for a food reward. Wandering shrews could not detect a plate smaller than 3.5×3.5 cm. Beyond 65 cm, shrews could not detect a 15×15 cm plate. Encountering a 4×4 cm plate with a hole (to mimic a tunnel opening), shrews detected accurately out to 30 cm. Buchler felt shrews do not use echolocation to detect prey or predators, but rather in a "search-image" mode to explore their environment. Shrews often chose the correct response but did not eat the mealworm. He believed that in these cases the open-arm of the Y-maze was selected because the opportunity to explore and, thus, "know" their environment was itself rewarding. Buchler (1976) noted that echolocation could be energetically more efficient than detailed tactile exploration of the environment, a particular advantage for species with a high metabolic rate, such as shrews. He also suggested that the use of low-amplitude pulses is adaptive for shrews concerned with short distance exploration and the need to avoid detection by predators.

Forsman and Malmquist (1988) studied echolocation abilities in *Sorex araneus* using a different apparatus: a central chamber from which six tubes (200 mm long, 16 mm diameter) radiated. The shrew's task was to select the single open tube for escape. Studies were conducted in the dark and in a soundproof box. Shrews produced ultrasonic sounds from 25 kHz to 95 kHz and discriminated the open tube between 30 s and 90 s. These authors thus concluded that this species has echolocation abilities.

Gould, Negus, and Novick (1964) monitored the per-

formance of short-tailed shrews (*Blarina brevicauda*) on the disk-platform apparatus and reported that these shrews produced clicks, puts, twitters, ultrasonic trains, chirps, and buzzes during the task (fig. 72.1). They clicked each time they extended their snout over the rim of a disk platform. He gave little data on durations or repetition rates and no data on peak frequency or bandwidth. He concluded that clicks were produced by forward movement of the tongue, puts and twitters as a consequence of sniffing, and chirps and buzzes by the larynx. Short-tailed shrew infants clicked as their mother approached to return them to the nest. Shrews frequently bobbed their head and twitched their snout while moving about in an unfamiliar environment; however, twitching of the snout was not always associated with audible sounds.

Gould (1969) noted that the sniffing rate of *Blarina* was close to the "putt" rate of another shrew, *Suncus*, and that putts could be the acoustic consequence of more intense smelling. Twitters were detected from short-tailed shrews during upward body extensions and "testing the air." According to Gould (1969), chirps, buzzes, and chirps graded into buzzes were used during agonistic encounters between short-tailed shrews. Acoustic information on the presence or absence of vegetative ground cover was suggested as an important adaptation for avoiding predation.

Tomasi (1979) conducted investigations on the echolocation abilities of short-tailed shrews trained to select open versus closed glass tubes. The open tube (imitating a tunnel) led to the shrews' home cage. No food reward was involved. Observations were conducted in the dark, under red lights, and olfactory cues were eliminated. Shrews could detect open versus closed tubes <61 cm long. Tomasi noted that glass tubes are highly reflective, so the detection distance probably was overestimated. When the diameter of a tube was varied, shrews discriminated diameters >0.63 cm. Even when the tube was bent at various angles from 0 to 90°, shrews detected the open versus the closed tube. Sound recordings made during some experiments showed the shrew produced at least one pulse before choosing which tube to enter. Pulses occurred singly, in pairs, or in triplets. Up to 20 pulses per series were given, with peak frequency between 30 kHz and 50 kHz. No waveforms or summary of acoustic parameters were published. Tomasi (1979) proposed that the primary function of echolocation in the short-tailed shrew is for exploring tunnels or avoiding obstacles, rather than for detecting prey.

In contrast, white-toothed shrews (*Crocidura russula*) in the Soricidae subfamily Crocidurinae produce clicks ranging from 10 kHz to 110 kHz, but Grunwald (1969) could not demonstrate echolocation.

Airapet'yants (1974) speculated about the usefulness of echolocation in shrews, given the low amplitude of the pulses and the cluttered vegetative environment they inhabit. He hypothesized that "in contrast to bats, which

make continual use of echolocation, shrews rely on it only sporadically . . . when an open area is crossed, echolocation could be an effective means for detecting the nearest shelters (stumps, bushes) . . . use of echolocation in unfamiliar terrain is especially important to small mammals which must conceal themselves from various predators in a suitable shelter." Airapet'iants proposed that water shrews (*Neomys fodiens*) sometimes migrate over land, and echolocation would be useful in that context. He further speculated that they could use echolocation to catch fish fingerlings that are especially active at night. Sales and Pye (1974) reported a variety of sounds between 5 kHz and 55 kHz made by the water shrew. In captivity, these shrews produced ultrasonic pulses before and after jumping from a platform to the floor of their cage. But because no sounds were detected when a shrew explored on the platform, the authors suggested that echolocation is unlikely.

Novick and Gould (1964) and Gould (1965) carried out intensive studies on echolocation in three insectivore species (*Hemicentetes semispinosus*, *Echinops telfari*, and *Microgale dobsoni*) in the family Tenrecidae (10 genera and 25 species), using the same disk-platform apparatus as in their shrew studies. Tenrecs produced clicks as they searched the platform and successfully found the platform in darkness, when blindfolded, and when open plastic tubes were placed in their ears. However, performance dropped sharply when their ears were plugged with cotton. The authors believed tenrecs echolocate by producing a series of high-amplitude, low-frequency clicks with their tongues (0.1–2.0 ms, from 9 kHz to 17 kHz). The duration of tenrec clicks was short (0.2–1.2 ms). Peak frequency in *Hemicentetes* clicks was 10–16 kHz. Clicks were produced singly, in doublets, or in triplets. When emitting clicks, *Echinops* licks its lips, moving the tongue laterally and upward above the lip, while *Hemicentetes* and *Microgale* simply open and close their mouths when clicking. Another tenrec, *Centetes*, also produce clicks this way. Amplitude of tenrec clicks varied, being very low intensity in *Centetes* and audible 7 m away in *Microgale*.

Tenrecs (fig. 72.2; table 72.3) produce a variety of sounds, including squeals, sniffs, twitters, and tongue-clicks. Kingdon (1974) and Nowak (1991) described tenrecs as having exceptionally small eyes, no auditory bullae, and an incomplete zygomatic arch. Sales and Pye (1974) reported sensitive hearing in the streaked tenrec (*Hemicentetes semispinosus*) from 2 kHz to 60 kHz and suggested that the tongue-clicks are produced at the best hearing sensitivities and stridulations sounds at the upper, less sensitive, part of their hearing range. Eisenberg (1975) reported that tenrec species, such as the streaked tenrec and the long-tailed tenrec (*Microgale longicaudata*), echolocate during foraging. The long-tailed tenrec resembles a shrew and occupies thick vegetation and ground litter in various habitats.

Novick and Gould (1964) described foraging by

groups of streaked tenrecs that have dorsal spines, which vibrate against each other to produce ultrasonics from 10 kHz to 70 kHz. The authors believed these sounds are related to communication rather than to echolocation.

Kingdon (1974) discussed two aquatic genera in the family Tenrecidae that forage along rivers in Africa and feed on aquatic invertebrates. These, the dwarf otter shrew (*Micropotamogale ruwenzorii*) and giant otter shrew (*Potamogale velox*), have several adaptations for aquatic life including long vibrissae, webbed feet, and long tails for steering. They have spectacular vibrissae that aid in underwater sensing, but also could function as a hydrofoil. The giant otter shrew feeds at night in the water and is thought to have very poor vision (Kingdon 1974). No data are available on sound production or possible echolocation abilities in the otter shrew, but its natural history is typical of mammals with echolocation abilities.

Sales and Pye (1974) reported that some hedgehogs (fig. 72.2) in the insectivore family, Erinaceidae (10 genera, 20 species), respond to ultrasonic whistles up to 84 kHz by jerking their bodies or turning their heads. Henson's (1961) studies on the anatomy of the middle ear of the European hedgehog, *Erinaceus europaeus*, indicated it hears ultrasonics. Another species of hedgehog, *Hemiechinus auritus*, responded to tones up to 40 kHz, but the most sensitive frequency was at 8 kHz (Sales and Pye 1974). Hedgehogs are terrestrial and nocturnal, with an omnivorous diet, feeding on invertebrates, small vertebrates, eggs, fruits, and carrion (Feldhamer et al. 1999). Hedgehogs have relatively small eyes and poor vision (Kingdon 1974). No studies on echolocation have been conducted on hedgehogs.

Moles (fig. 72.2; table 72.3) in the insectivore family Talpidae (17 genera, approximately 42 species) eat earthworms and soil invertebrates. Moles live in semipermanent underground tunnels and feed day and night in their tunnels. According to Gorman and Stone (1990), talpids have "poor vision, acute hearing, and exquisite sense of touch." The pinnae are reduced or absent and the eyes are minute (Feldhamer et al. 1999). Mole snouts are tipped with thousands of tactile-sensing Eimer's organs. The star-nosed mole (*Condylura cristata*) exhibits a ring of tentacle-like sensing organs on the snout. Hairs on the forefeet and tail also have a sensory ability in some moles. Gorman and Stone (1990) concluded that moles rely primarily on tactile senses for navigation and foraging. Although they should be, studies designed to test echolocation abilities have not yet been conducted on true moles.

The family Talpidae also includes desmans and shrew-moles (table 72.3). Desmans live near streams or rivers, have a diet of aquatic invertebrates and fish, and are good swimmers. Richard (1973) reported that the Iberian desman echolocates and uses its long flexible snout to maneuver and locate prey under water. Again, desmans are a good candidate for studies on echolocation.

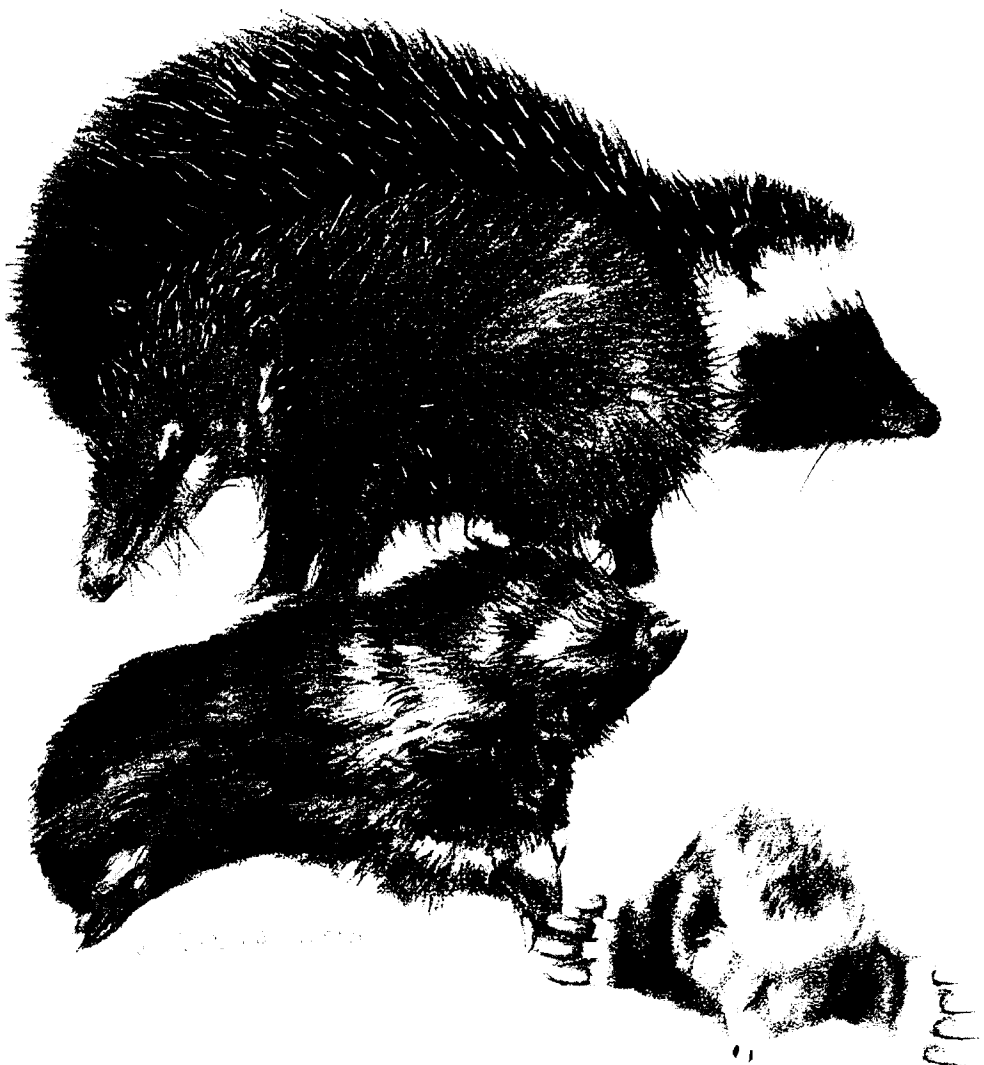


Fig. 72.2. Representative fossorial mammals in the order Insectivora. From top to bottom – A: hedgehog (*Ernaceus algurus*); B: tenrec (*Tenrec ecaudatus*); C: golden mole (*Chrysochloris* spp.); D: eastern mole (*Scalopus aquaticus*). Drawing by Jennifer Webster.

No information is available on the hearing, sound-production, or echolocation abilities of African golden moles (fig. 72.2, table 72.4) in the insectivore family Chrysochloridae (7 genera, 19 species). Golden moles have no pinnae, large auditory bullae, and an “unusual hearing apparatus” (Kingdon 1974). Their eyes are vestigial and usually covered by skin (Kingdon 1974). Eyes of the golden mole remain closed at birth and eventually are covered with fur (Gorman and Stone 1990). Golden moles burrow deep and, despite their lack of external eyes, have an excellent sense of direction. They feed on earthworms, beetles, grubs, slugs, and snails. In the presence of a worm, they respond with movements of the head and twitching of the snout, “as if to get a sense of direction” of the prey, but they are indifferent to the smell and sounds of insect prey (Kingdon 1974). Golden moles are active during the day or night, but they remain generally silent. Golden moles are thus also a good candidate for echolocation studies.

Little is known about sound production in the insectivore family Solenodontidae (table 72.3), the Cuban and Haitian solenodonts (2 extant species). Solenodonts

have long pinnae, small eyes, and a sensitive snout used in capturing prey (Feldhamer et al. 1999). They live in rocky, brushy, or forested areas and are omnivorous. Eisenberg and Gould (1966) recorded broadband clicks from *Solenodon paradoxus* that were 0.1–3.6 ms, between 9 kHz and 31 kHz, and produced in bursts of one to six clicks when the animal explored a new area or encountered a stranger. No controlled echolocation studies have been conducted in this family of insectivores.

New Data on Echolocation in Shrews

In the wild, short-tailed shrews (fig. 72.1) inhabit woodland or damp areas of central and eastern North America, with leaf litter or humus ground cover. It is a scansorial species foraging on worms, snails, slugs, and insect larvae (Rood 1958). Using an Anabat II bat detector set with a division rate of eight (i.e., monitoring up to 160 kHz), we recorded vocalizations from two adult short-tailed shrews (*Blarina brevicauda*) on a Marantz cassette recorder (fig. 72.3). Shrews were tested individually in the dark in a 20 gallon aquarium and in a closed

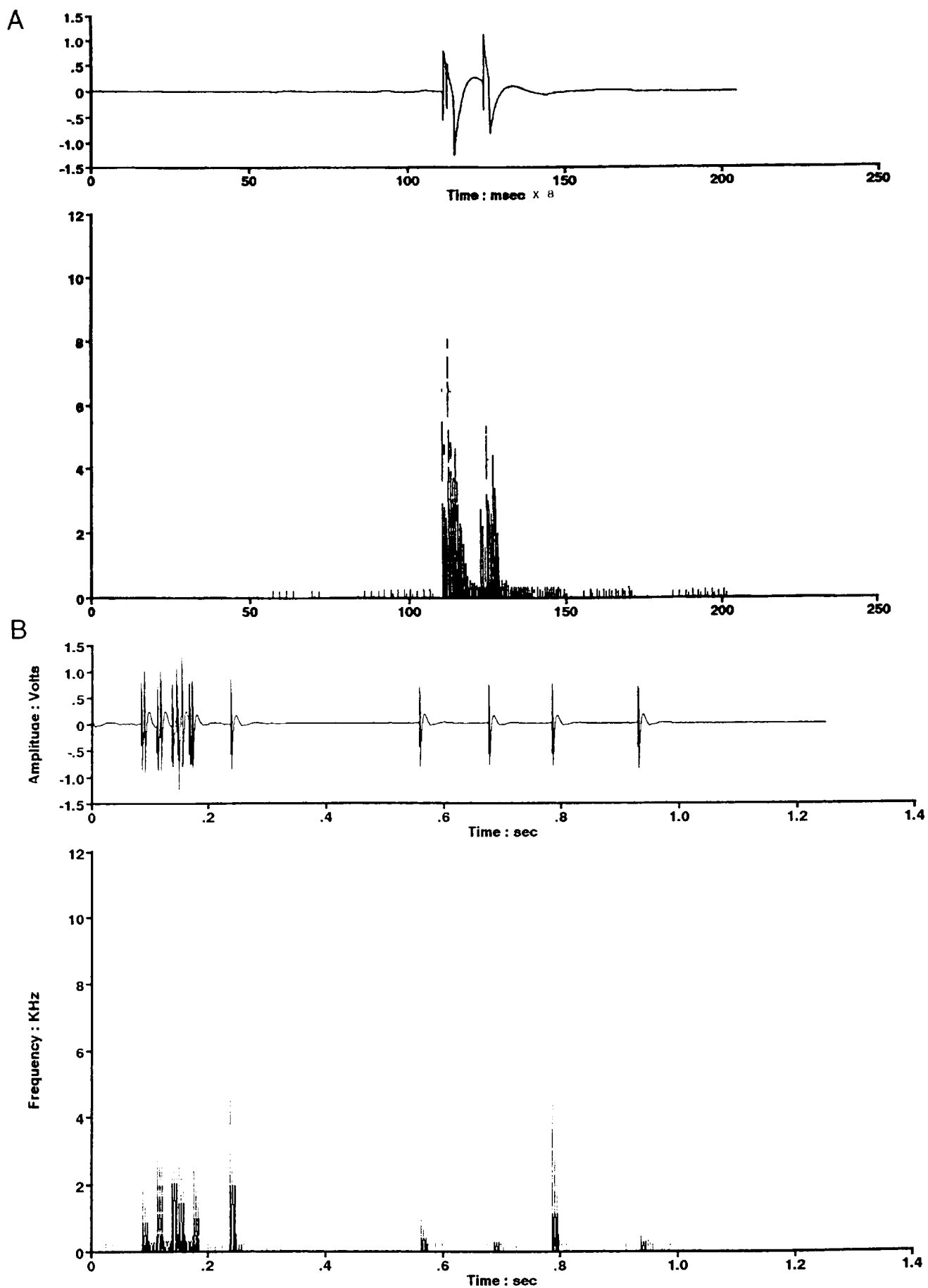


Fig. 72.3. Oscillograms and sonograms of (A) pulses and (B) trains produced by the short-tailed shrew in a dark, soundproof box

box lined with neoprene. Shrews bobbed their heads and twitched their snouts while producing ultrasonic pulses. Sonograms of clicks duplicated those reported by Gould (1969, fig. 2H). However, we did not record putts, twitters, chirps, or buzzes. Clicks were produced singly or in trains. Sometimes clicks were in pairs, but there was no phase shift to indicate the second pulse was an echo. The peak frequency (in fig. 72.3 this translates into 30 kHz to 50 kHz) was the same noted by Tomasi (1979) and by Gould (1969) for this species.

Least shrews, *Cryptotis parva* (fig. 72.1), are nocturnal and typically inhabit woodlands and open fields of the eastern North and Central America (Whitaker 1974). This shrew either uses the runways of other small mammals or requires the presence of dense herbaceous vegetation for cover. It forages on surface-dwelling invertebrates and probes among vegetation or leaf litter, but it does not actively dig for prey (Churchfield 1990).

Highly social, least shrews huddle together in the wild and in captivity and often produce clicks that have "a high localization valence" (Gould 1969). They readily produce low-amplitude clicks accompanied by head bobbing and snout twitching. *Cryptotis* produce putts and twitters, chirps, and buzzes, but mostly clicks (Springer 1937; Moore 1943; Gould 1969). One sound grades into another—that is, a chirp into a buzz and a putt into twitter. Perhaps because of equipment limitations, little data

on sound duration or repetition rates are reported, and no data on peak frequency or bandwidth are available. Gould (1969) reported that clicks were contact calls used among individuals and produced by young pups at only nine days old. Both adult males and females produced clicks during solitary exploration, courtship, mouth washing, investigation of a small object, and when the snout was extended over the edge of an object. The prominence of clicks in the least shrew's repertoire compared to other sorcids reflects this shrew's social nature.

We tested eight adult least shrews individually in a series of 5 min orientation and foraging tasks on an experimental board. The carpet-covered experimental board (27 × 54 cm) had an inner and outer box of equal area (fig. 72.4). Orientation tasks included open board, presence of Plexiglas walls around outer edge of board, and the presence of a Plexiglas tube in the center of the board. Foraging contexts included no mealworm present, mealworm moving freely on the board, and a mealworm pinned in center of the board. Trials were run in ambient daylight and again in a dark room. Behaviors and sounds from shrews were recorded using a video or night vision camcorder (Sony model CCD-TRV815) and a Marantz (model PMD-420) cassette recorder. The receiving beam pattern of an Anabat II bat detector (division rate of four or a monitoring range up to 80 kHz)

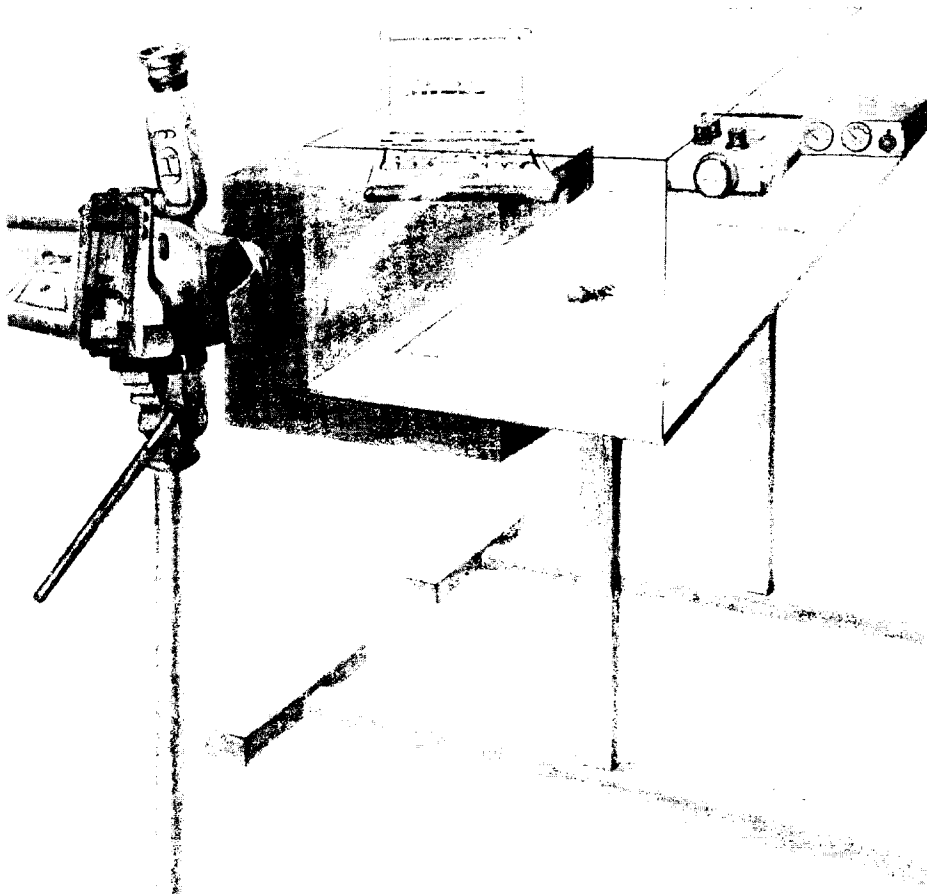


Fig. 72.4. The experimental board, bat detector, video camera, and computer with software used to examine echolocation abilities of the least shrew. Note that the area in the outer box equals the area in the inner box. Plexiglas walls surround the board with a shrew in the middle. Drawing by Jennifer Webster.

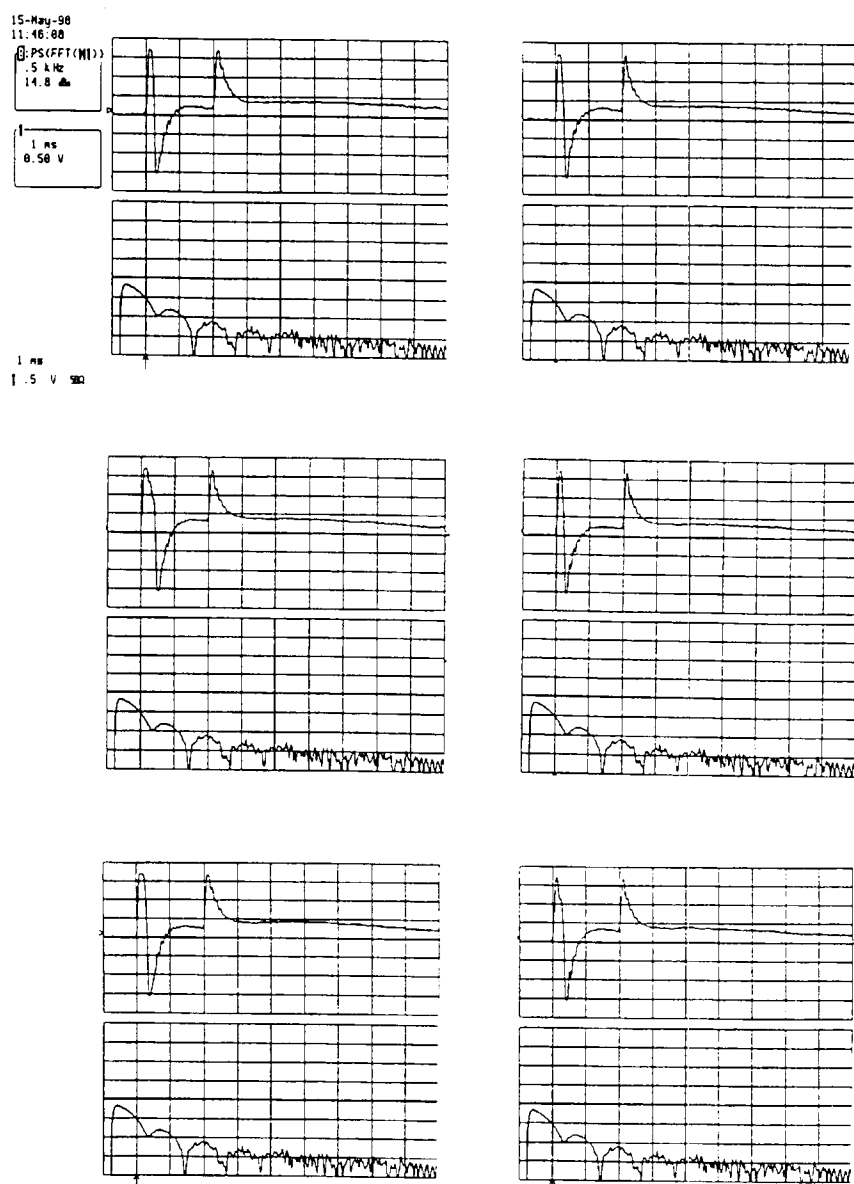


Fig. 72.5. Oscillograms and power spectra of representative pulses produced by the least shrew during various orientation and foraging tasks on the experimental board

covered the entire board. Preliminary tests indicated that a higher division rate was not necessary. Sounds were monitored using Audioscope real-time sonogram software on a laptop computer (Gateway Solo 2000). Data transcribed from video and/or cassette tapes for each pulse included whether the shrew was in the inner or outer box, the number of pulses in a train, whether the shrew searched the board, the number of times the shrew ate a worm, and total time spent eating.

The only sounds detected from least shrews during the orientation/foraging tasks were clicks, although we heard twitters and putts when the shrews were in their home cage. Least shrews showed little variation in waveform and power spectra of pulses among individuals. Typical clicks (fig. 72.5) had a doublet waveform (Buchler [1976] reported similar double pulses in *Sorex vagrans*). Pulses by least shrews were low in amplitude (ac-

tually audible when held close to the ear). We expected pulses would be ultrasonic, as in other shrews, but in fact pulses were between 200 and 12000 Hz, with the average peak frequencies between 242 and 690 Hz (table 72.6). Gould (1969) also reported the frequency range of low-intensity sounds from least shrews as between 500 and 1200 Hz and detectable 5 cm away. In retrospect, the bat detector simply amplified signals and was unnecessary for detecting ultrasonics in this species.

All shrews produced pulses at the same rate and foraged and oriented in the same manner during light and dark trials, confirming the least shrews' poor vision. The mean number of pulses was higher during orientation tasks than foraging (table 72.6). The highest number of pulses per trial (table 72.6) was produced during the wall-present trials, compared to the least number during free-moving worm trials. The mean peak frequency of

TABLE 72.6. Summary of individual pulse characteristics for least shrews ($n = 8$ shrews).

Trial condition	Task type	Mean number of pulses per trial	Mean pulse peak frequency (Hz)	Mean pulse bandwidth (Hz at +3dB)	Mean pulse duration (ms)
Open board	orient	10.0	242	271	65.8
Wall	orient	20.2	308	285	79.4
Tube	orient	16.3	564	489	42.5
Worm pinned	forage	7.3	578	509	29.8
Worm free	forage	4.0	690	702	19.2

TABLE 72.7. Summary of pulse train characteristics for least shrews ($n = 8$ shrews).

Trial condition	Task type	Mean number of pulses per train	Mean "train" duration (ms)
Open board	orient	2.7	0.3
Wall	orient	2.5	0.3
Tube	orient	3.0	0.5
Worm pinned	forage	13.8	2.8
Worm free	forage	12.9	2.6

TABLE 72.8. Summary of time spent by least shrews in different locations and tasks on the experimental board ($n = 8$ shrews).

Trial condition	Task type	Inner box (s)	Outer box (s)	Mean eating time (s)	Mean searching time (s)
Open board	orient	71.6	224.1	N/A	300.0
Wall	orient	41.3	259.5	N/A	300.0
Tube	orient	170.4	109.7	N/A	300.0
Worm pinned	forage	179.3	120.6	129.7	169.5
Worm free	forage	177.2	117.1	128.3	172.0

pulses was lowest during open-board trials, higher during orientation tasks (wall and tube trials), and highest during foraging tasks (pinned and free worm trials). Bandwidth and mean pulse duration showed a similar progression. These patterns make sense in that a shrew can increase the rate of pulses for more detailed discriminations when investigating its environment.

The largest number of pulses per train and the longest trains were produced during actual consumption of the worm (table 72.7). During orientation tasks, the pulses typically were two to three per train and 0.2–0.5 ms. On an open board, shrews spent more time in the outer box, searching the perimeter, but produced few pulses (tables 72.6 and 72.8). When walls were present, there was a significantly higher number of pulses in the outer box ($F = 64.5, p < 0.01$). When the tube was present, shrews produced a significantly higher number of pulses in the inner box containing the tube ($F = 67.7, p < 0.02$) and spent less time in the outer box. When no worm was present, shrews produced more pulses at all locations on the board and had a significantly longer

search time ($F = 564.5, p < 0.01$). As the number of pulses increased, so did search time ($r = 0.31$) and the number of times the shrew successfully found ($r = 0.35$) and successfully ate the worm ($r = 0.36$). Collectively, these experiments showed that least shrews produced pulse trains in both foraging and orientation tasks.

Discussion of Echolocation in Insectivores

According to Gould (1969), there is no way to distinguish between pulses used for communication and those used for echolocation. Bateson (1966) noted that these two contexts are not necessarily exclusive. A pulse produced by a shrew often has a sonic portion (a "ticklaut") associated with an ultrasonic sound (Dijkgraaf 1932). Pulses in shrews "may owe their shortness and intensity to the fact that they are a lower variant of supersonic pulses used in echolocation and would have an immediate value if used in echolocation of the source, as well as a potential value in communication, both functions may have evolved together in Insectivora" (Andrew 1964, pp. 287–288). Similarly, "calls emitted during an early period of an animal's life history may occur later in life in a different context" (Andrew 1964, p. 231). Pulses emitted by shrews about to suckle, during caravan formation, and separation from the mother occur in adults during exploration and at increased rates in courting males. Gould (1969) noted that an infant *Cryptotis* emitted twitters when a dispersed group of four was reunited with their mother in the nest. The putt (300–1000 Hz, 0.6–0.9 s, repeated 10–14 per s) was produced by both sexes of *Cryptotis* and detectable only 5 cm away Gould (1969). Putts were used when a lone shrew explored a strange situation (Gould 1969). *Cryptotis* usually emitted twitters during horizontal movement and by adults when infants were returning to the nest. Twitters probably were used at greater interanimal distances than puts. Ultrasonic trains of pulses (50–420 ms, typically 12 pulses in 162 ms) were produced by least shrews during intense exploration, and Gould (1969) felt they were used for echolocation. High-intensity buzzes were used in close encounters between shrews.

The amplitude of shrew pulses is much lower than that reported in echolocating bats. According to Churchfield (1990), this is adaptive, because "if shrews were to

use high-intensity emissions in their cluttered habitats of plant stems, roots, logs, and rocks they would be inundated with echoes from every conceivable surface. So high-frequency, but low-intensity pulses are probably more useful. These low-intensity ultrasounds attenuate rapidly and are scattered easily by small objects. This means that predators, even if they can detect ultrasounds, would have difficulty locating the source. So echolocation in shrews may be a means of exploring the habitat without attracting the attention of predators."

Noting the very different tooth structure in shrews compared to other mammals (fig. 72.6), we wonder whether the first incisor functions in sound production. In red-toothed shrews (i.e., subfamily Soricinae), all teeth are tipped with a dark red coloration, an oddity among mammals. The pigment is a form of iron. Shrews have considerable tooth wear and the red coloration could be deposited to strengthen the tooth (Churchfield 1990). In addition, soricids have an exceptionally large, bicuspid first upper incisor, and a large, procumbent first lower incisor, both rarities among mammals. The upper and lower first incisors work together as tweezers to pick up insect prey (Feldchamer et al. 1999). Kingdon (1974) suggested that some shrews "crack their teeth together and make a number of squeaks and chatters." The first incisors are larger than all other teeth, and we wonder whether they provide a large surface area against which the tongue (or lower incisors) strikes to produce a click. The notable tooth wear reported in shrews could be explained if the teeth are used for both eating and pulse production. We further speculate that the double structure of the pulses could result from striking of the two cusps of the upper first incisor. Some odontocetes in the family Phocoenoidea (*Phocoenoides* and *Ponteporia*) also produce double echolocation pulses. The situations under which a mammal produces a single or double pulse for echolocation should be investigated.

Previous Studies on Echolocation in Rodents

Airapet'yants and Konstantinov (1970) first summarized studies on possible echolocation in rodents. The dormouse (*Glis glis*) found food on a slim perch at 40 cm away in utter darkness by jumping onto it, even though the location was changed many times. Kahmann and Ostermann (1951) conducted a series of experiments in the dark with golden hamsters and redbacked voles in which normal and blind animals needed to jump from a table to a platform below leading to a food reward. The vertical and horizontal position and distance of the platform relative to the table was varied. After just a few trials, performance was high. Golden hamsters had a critical distance of 6–8 cm and redbacked voles from 10 to 14 cm. Although the authors monitored up to 70 kHz, ultrasonic sounds were not detected during the trials. Removing the vibrissae from blind redbacked voles did not affect their performance. However, blind red-

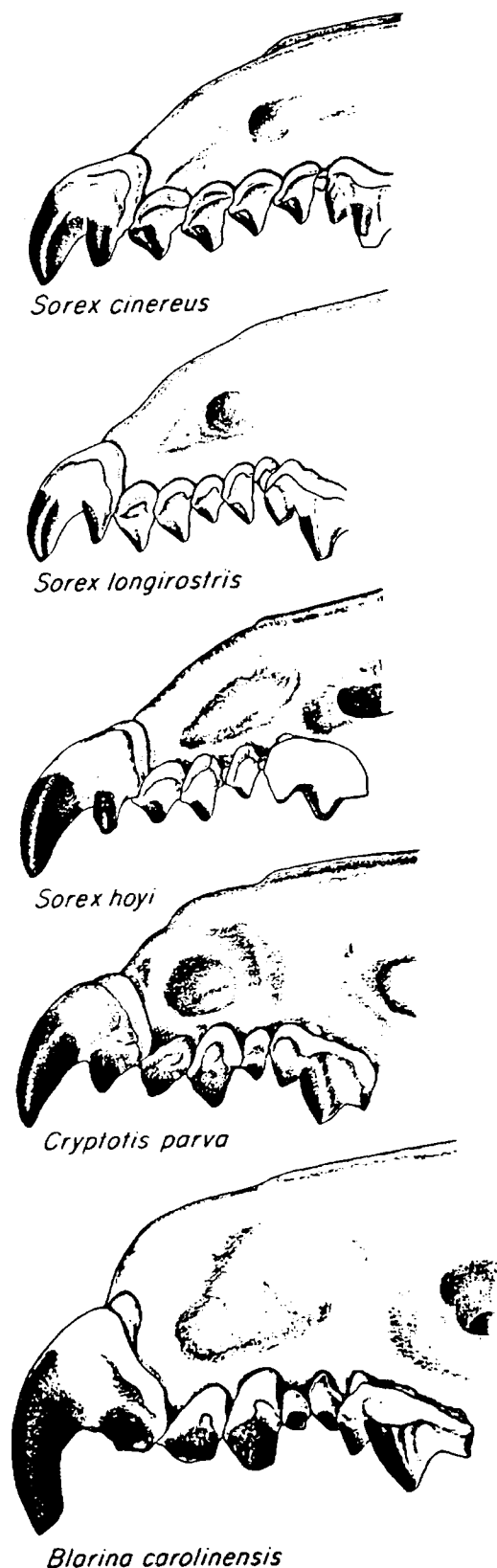


Fig. 72.6. Lateral view of the rostrum of five shrews. Note the prominent bicuspid first incisor and dark coloration on teeth (from Hoffmeister 1989).

backed voles with plugged ears were not able to perform and jumped randomly from the tabletop. The authors concluded that these species shifted to alternate senses when blind, but they could not conclusively prove echolocation.

In contrast, Airapet'yants (1974) tested three blind dormouse species in the dark with a similar table/platform apparatus used by Kahmann and Osterman (1951). None of the dormice could find the food except when they were close enough to use their vibrissae.

Airapet'yants (1974) conducted another study on six golden hamsters using two platforms; the lower one could be adjusted in a vertical and horizontal position to the upper platform. The hamster had to jump from the highest platform to the lower platform to receive food. After initial training, the hamsters were blinded. The hamsters ran around the edge of the upper platform, hanging their body over the side to explore below, then raising back onto the top platform. Faint "sniffing" sounds were heard when they suspended themselves, but a single attempt to monitor at 70 kHz produced no ultrasonics. By the third experiment, hamsters were performing at 70% of baseline trials. When the vibrissae were removed, hamsters could not perform the task. The authors concluded touch, rather than echolocation, was needed for the task.

Airapet'yants (1974) conducted studies on the voles *Microtus socialis* and *Microtus gregalis* with the identical setup to the hamster study; however, the results were different. Blind voles of these two species readily found the lower platform when it was <4 cm away. In contrast to hamsters, blind voles without vibrissae did not have impaired orientation to the lower task. The platform was washed between trials to eliminate olfactory cues, but performance continued to be high. In further studies, the blind voles with partially blocked hearing or perforated eardrums could no longer perform the task. The authors did not hear sounds associated with jumps and did not have equipment to record during the experiments. They believed they confirmed echolocation in these two species. However, they conducted another study with the bank vole (*Clethrionomys* spp.) using a platform from which one of eight paths radiated to a food reward. This species when blind conducted the task successfully, but they failed when vibrissae were removed.

Three sets of experiment were published on echolocation on blind laboratory rats (*Rattus norvegicus*) orienting in a Y-maze. In these experiments, one arm of the maze was obstructed, and rats were trained to find and move through the remaining open arm to find food. With training, rats were successful in 75–80% of trials. Authors concluded that hearing was involved in successful orientation, because if the ears were blocked (Rosenzweig, Riley, and Krech 1955; Chase 1980), or the hearing temporarily masked by loud white noise for seven days (Bell, Noble, and Daves 1971), success dropped to the level of chance. When the plate was moved so it was

>23 cm from the rat (Bell, Noble, and Daves 1971) or the angle of the plate was changed to 45° (Rosenzweig, Riley, and Krech 1975), success dropped to chance levels, indicating the placement of the plate was important, as it would be if rats were bouncing acoustic signals off it. These reports did not indicate whether they controlled for the odor of the food reward. Rosenzweig, Riley, and Krech (1955) noted a "variety of other auditory cues, such as foot scratching, stepping, teeth clicking, sniffing, and sneezing could not be ruled-out as possible passive auditory cues for the task."

Chase (1980), using laboratory rats in a Y-maze, reported a correlation between ultrasonic vocalizations and correct detection. Chase (1980) only published an oscillogram analysis, so characteristics of rat echolocation signals are minimal: four to eight low-amplitude pulses per series, with most energy between 40 kHz and 50 kHz. Pulses often were in pairs.

Collectively, these studies on echolocation in rodents paint a mixed picture. A more careful selection of study animals based on the ecology of the species and the aforementioned profile of correlates with echolocation could provide a better understanding of senses used in orientation in rodents.

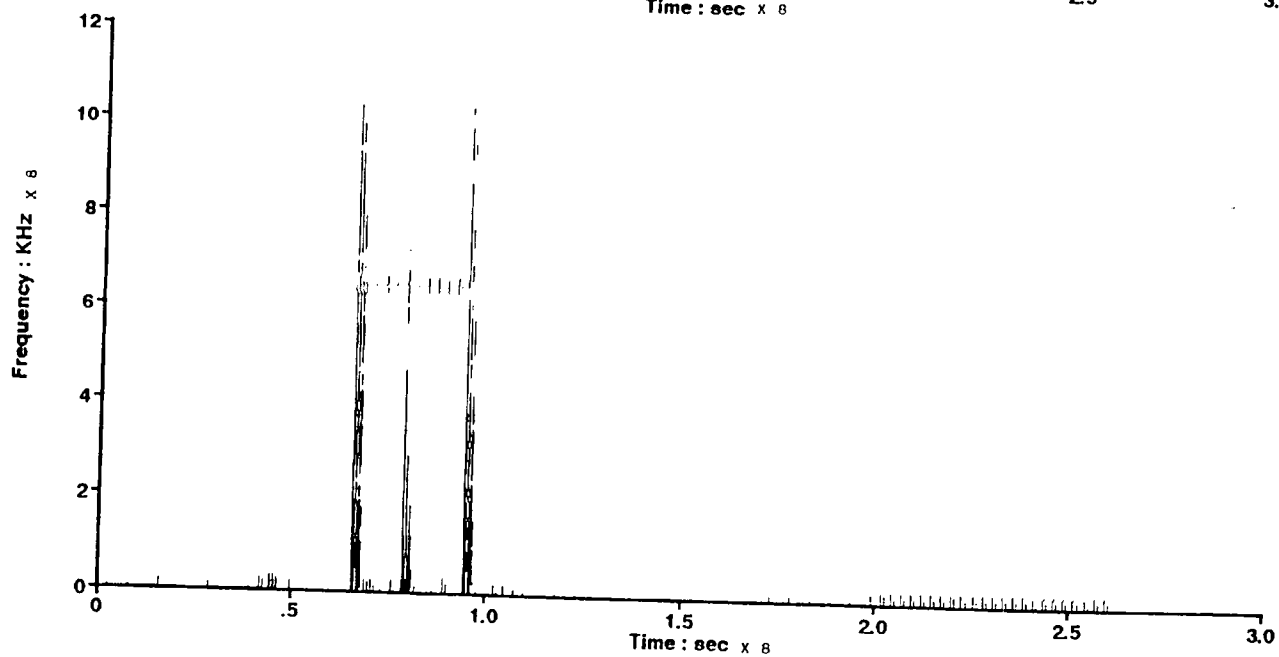
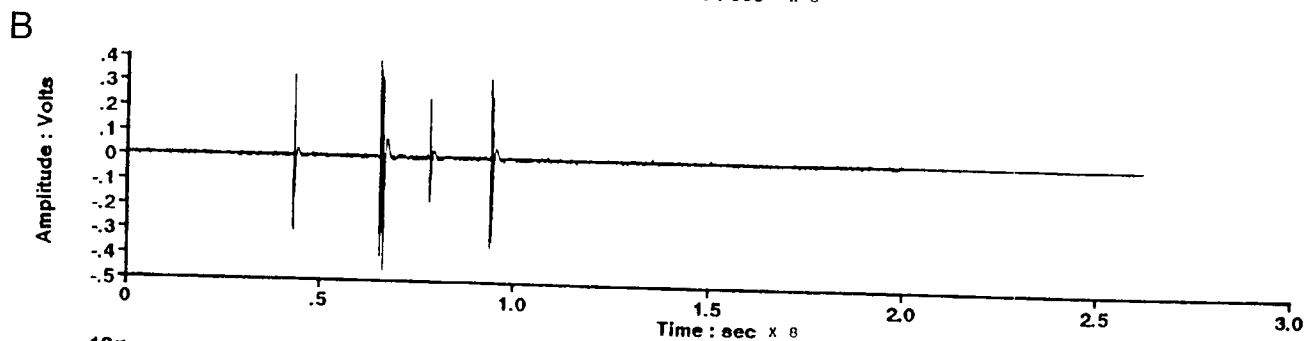
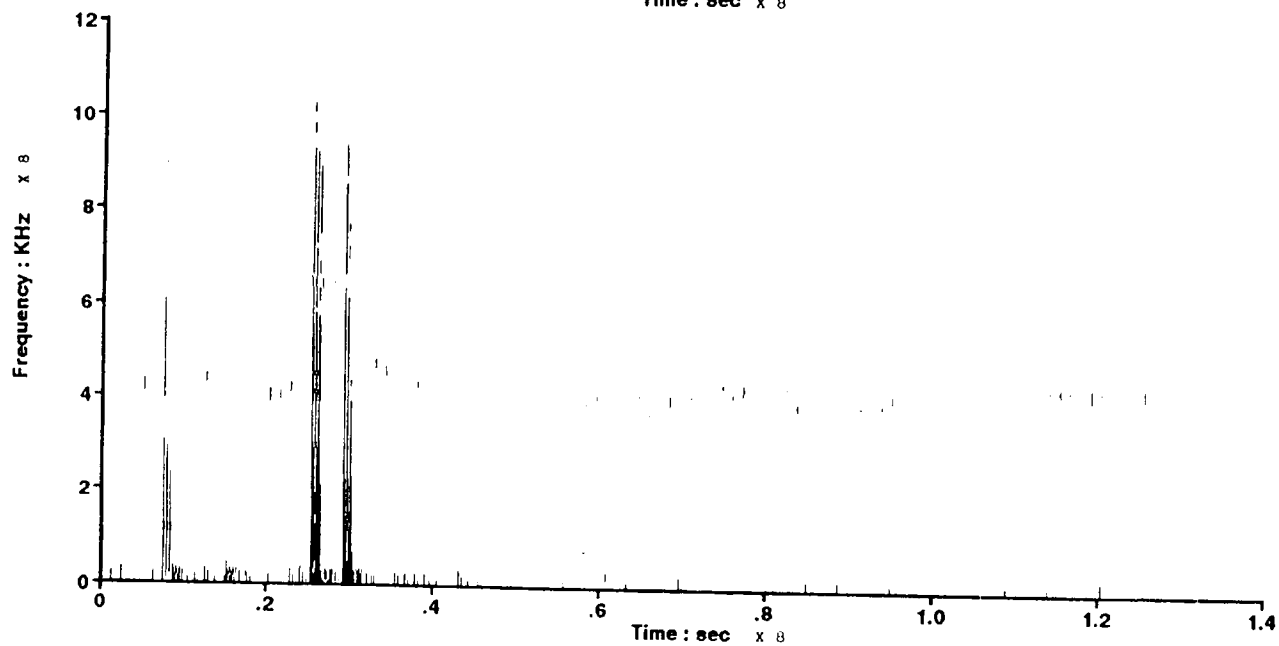
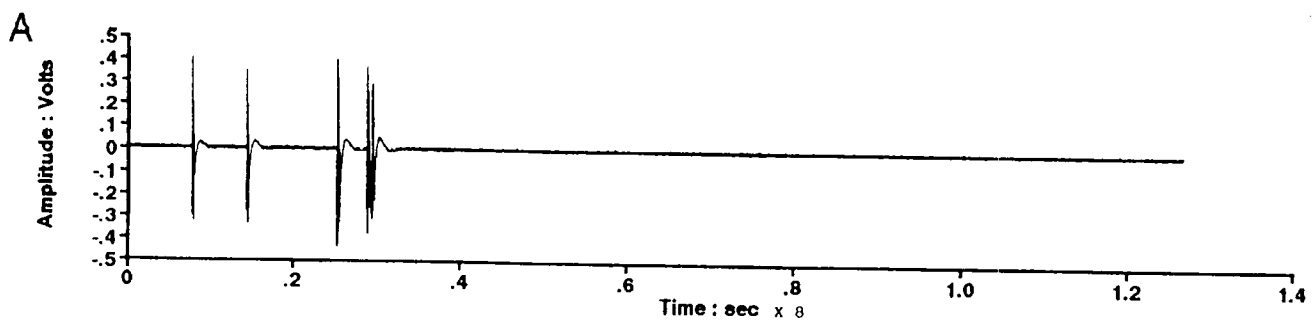
New Data on Echolocation in Rats

We made broadband recordings of sounds from laboratory rats in daylight and darkness. Using an Anabat II detector set at a division rate of eight (i.e., monitoring up to 160 kHz), we made 5 min recordings on a Marantz cassette recorder of five adult laboratory rats in their cages in ambient light. Rats produced ultrasonic sounds, but none were pulses. When each rat was enclosed in a dark, neoprene-lined box (to prevent passive acoustic cues), several single and trains of pulses were produced with waveforms reminiscent of echolocation pulses (fig. 72.7). The fact that only isolated rats in the dark produced these pulses gives further support to echolocation abilities.

Discussion of Echolocation in Insectivores and Rodents

Mammals use many types of sensory cues to monitor their environments, navigate, and find food. In addition to toothed bats and dolphins, echolocation abilities in mammals such as insectivores and rodents merit further investigations. Touch is probably a very important sense in mammals living in limited light environments. Most have vibrissae and many have specialized tactile organs (such as the Eimer's organ in moles or the elaborate vibrissae of otter shrews), but most studies on echoloca-

Fig. 72.7. (opposite) Oscillograms and sonograms of (A) pulses and (B) trains produced by a laboratory rat in a dark, sound-proof box



tion have not controlled adequately for possible touch or olfactory senses. Special consideration should be given to nocturnal, semifossorial, fossorial, or visually impaired mammals, and to species with broadband hearing or ultrasonic vocalizations. Echolocation seems to occur in many insectivore families examined to date. More research is needed on signal characteristics, detection, and discrimination abilities in all groups of echolocating insectivores.

One might speculate that because shrews, rats, and bats echolocate in air, rather than in water like dolphins, their signals would be similar to those of bats. Interestingly, the echolocation signal of shrews and the laboratory rat is a single or double broadband, short pulse like the signal of echolocating dolphins, rather than the longer FM, CF, CF-FM signals of bats. This could relate to the close ranges at which shrews would use echolocation. Studies on aquatic insectivores (Sergeev 1973), like water shrews (*Sorex palustris* or *Neomys fodiens*), desmans, or otter shrews, or on aquatic rodents (in the sub-

family Ichthyomyinae—especially *Chibchanomys*, *Anatomyomys*, *Ichthyomys*, and *Rheomys*) that spend significant amounts of time foraging under water, might provide some interesting data on signal type versus medium. Fortunately, technology now is available to collect and to analyze short ultrasonic sounds often associated with echolocation in mammals.

Acknowledgments

The authors thank Bob Connour and Jaime Welling for their assistance in making recordings of shrews and rats. We thank the John G. Shedd Aquarium for use of the Bioacoustics Laboratory and the Biology Department of Western Illinois University for the acquisition of equipment. Paul Anderson, Bob Timm, and Barbara Blake provided very useful comments on this manuscript, and Jennifer Webster created the artwork for this chapter.

{ 73 }

Potential Use of Low-Frequency Sounds by Baleen Whales for Probing the Environment: Evidence from Models and Empirical Measurements

Christopher W. Clark and William T. Ellison

Introduction

Over the course of evolution organisms change or adapt through the processes of natural and sexual selection and genetic drift (Gould and Lewontin 1979). Natural variability within a population is the fodder for selection, and an organism with traits best adapted to its environment is most likely to transfer genes to the next generation. By this mechanism, heritable traits are selected for and become common throughout a population. The time scales over which selected traits become common in a population are dependent on a number of internal and external factors. Internal factors include such things as physiology, morphology, and phylogeny, while external factors primarily include the ecological, social, and physical environments. Generally, fixation of an adapted trait is thought to take many thousands of generations. However, recent observations of natural se-

lection in birds, fish, and insects indicate that in some cases selection, as manifested in the frequency distributions of phenotypic traits, can begin to occur within as short a period as several generations (Schluter and McPhail 1992).

Behavioral traits are subject to many of the same selective mechanisms, but not necessarily the same selective pressures as nonbehavioral (e.g., morphological) traits, and are generally more labile than morphological characteristics. In this discussion of mysticete acoustics, we take the simplistic view that characteristics shared broadly across multiple species are considered more ancestral than features shared only at the species level. We introduce this conceptual framework as the basis for gauging the degree and approximate order of the physical environment's effect on the features of sounds within a specialized group of marine mammals, the baleen whales (mysticetes).